

BIOCHEMISTRY 2017 EXAM

1. Which one of the following causes free radicals to form, potentially putting you at greater risk of heart disease?

- A. High-fat foods
- B. Taking high amounts of a single antioxidant
- C. Exercise
- D. All of the above
- E. None of the above

Correct Answer: D. All of the above

Biochemical Rationale:

Free radicals are generated by various physiological and environmental factors. High-fat foods can increase oxidative stress through lipid peroxidation. Excessive intake of a single antioxidant can paradoxically become pro-oxidant. Exercise, while generally beneficial, increases oxygen consumption and can transiently elevate free radical production. All of these factors can contribute to oxidative stress and increased cardiovascular risk.

2. Which protein is different regarding its biological function in the body?

- A. Prion
- B. Ricin
- C. Snake venom toxin
- D. Thrombin

Correct Answer: D. Thrombin

Biochemical Rationale:

Thrombin is a physiological serine protease essential for blood coagulation, converting fibrinogen to fibrin. In contrast, prions are misfolded infectious proteins, ricin is a toxic lectin from castor beans, and snake venom toxins are harmful exogenous proteins. Thrombin serves a normal biological function, while the others are pathological or toxic agents.

3. Which of the following is a regulatory protein?

- A. Calmodulin
- B. Insulin
- C. p53
- D. All of the above

Correct Answer: D. All of the above

Biochemical Rationale:

Calmodulin is a calcium-binding regulatory protein that modulates many enzymes. Insulin is a peptide hormone that regulates glucose metabolism. p53 is a transcription factor that regulates cell cycle checkpoints and apoptosis. All three function in regulatory capacities.

4. Which of the following biomolecule proteins is the source of light in fireflies?

- A. Lucinase
- B. Luciferin

- C. Luciferase
- D. Photoprotein

Correct Answer: B. Luciferin

Biochemical Rationale:

Luciferin is the substrate (a small molecule) that emits light when oxidized by the enzyme luciferase in the presence of ATP and oxygen. Luciferase is the enzyme catalyst, while luciferin is the actual light-emitting biomolecule.

5. Which of the following plasma proteins is involved in preventing Fenton reaction (iron/copper catalyzed free radical damage)?

- A. Ceruloplasmin
- B. Haptoglobin
- C. Transferrin
- D. Ferritin

Correct Answer: A. Ceruloplasmin

Biochemical Rationale:

Ceruloplasmin is a copper-containing plasma protein with ferroxidase activity that oxidizes ferrous iron (Fe^{2+}) to ferric iron (Fe^{3+}), preventing Fe^{2+} from participating in the Fenton reaction, which generates highly damaging hydroxyl radicals. It also binds copper, preventing copper-catalyzed free radical formation.

6. Which protein (enzyme) is the first to cut in Base Excision Repair (BER) mechanism?

- A. DNA glycosylase
- B. AP endonuclease
- C. Phosphodiesterase
- D. DNA ligase

Correct Answer: A. DNA glycosylase

Biochemical Rationale:

DNA glycosylase initiates BER by recognizing and removing damaged or inappropriate bases (e.g., uracil, oxidized bases) through hydrolysis of the N-glycosidic bond, creating an apurinic/aprimidinic (AP) site. AP endonuclease then acts subsequently to nick the backbone.

7. Which of the following is NOT true about the human genome?

- A. Nuclear genome takes the "lion's share" of total DNA
- B. Most of mitochondrial genome is non-coding ("idle")
- C. Mitochondrial DNA is circular and maternally inherited
- D. Nuclear DNA contains both exons and introns

Correct Answer: B. Most of mitochondrial genome is non-coding ("idle")

Biochemical Rationale:

The human mitochondrial genome is highly compact and efficient, containing 37 genes (13 protein-coding, 22 tRNA, 2 rRNA) with virtually no introns or non-coding sequences. In contrast, the nuclear genome contains large amounts of non-coding DNA, including introns and intergenic regions.

8. Most amino acids are coded by more than one codon, except:

- A. Threonine and Tyrosine
- B. Glycine and Serine
- C. Methionine and Tryptophan
- D. Alanine and Valine

Correct Answer: C. Methionine and Tryptophan

Biochemical Rationale:

The genetic code is degenerate, meaning most amino acids are encoded by multiple codons. However, methionine (AUG) and tryptophan (UGG) are each encoded by only a single codon. AUG also serves as the start codon for translation initiation.

9. Which of the following is NOT a disorder linked to defective DNA repair mechanisms?

- A. Xeroderma pigmentosum
- B. Hereditary nonpolyposis colon cancer (HNPCC)
- C. Fanconi's anemia
- D. Phenylketonuria

Correct Answer: D. Phenylketonuria

Biochemical Rationale:

Phenylketonuria (PKU) is caused by a deficiency of phenylalanine hydroxylase, leading to accumulation of phenylalanine. It is a metabolic disorder, not a DNA repair defect. Xeroderma pigmentosum (NER defect), HNPCC (mismatch repair defect), and Fanconi anemia (crosslink repair defect) all involve defective DNA repair.

10. A common clinical feature seen in persons affected with a disorder of DNA repair is:

- A. Albinism
- B. Predisposition to cancer
- C. Excess stability of the genome
- D. Hyperproduction of immunoglobulins

Correct Answer: B. Predisposition to cancer

Biochemical Rationale:

Defective DNA repair leads to accumulation of mutations, genomic instability, and increased susceptibility to cancer. Examples include xeroderma pigmentosum (skin cancer), HNPCC (colorectal cancer), and Li-Fraumeni syndrome (multiple cancers).

11. Paternal relationship between a man and infant can best be determined by DNA fingerprinting. Which of the following sequences is most conveniently analyzed?

- A. Histocompatibility loci (HLA genes)
- B. Centromeres
- C. Microsatellite tandem repeats (STRs)
- D. Restriction enzyme recognition sites

Correct Answer: C. Microsatellite tandem repeats (STRs)

Biochemical Rationale:

Short tandem repeats (STRs) are highly polymorphic microsatellite sequences (2–7 bp repeats)

that vary in repeat number among individuals. They are the standard markers for DNA fingerprinting in paternity testing and forensic analysis due to their high variability and ease of amplification by PCR.

12. Which of the following vitamins/enzymes is unable to protect against free radical damage?

- A. β -Carotene
- B. Glutathione peroxidase
- C. Vitamin B6
- D. Vitamin E

Correct Answer: C. Vitamin B6

Biochemical Rationale:

Vitamin B6 (pyridoxal phosphate) serves as a coenzyme in amino acid metabolism (transamination, decarboxylation) but does not function as an antioxidant. Vitamin E (lipid-soluble antioxidant), β -carotene (singlet oxygen quencher), and glutathione peroxidase (enzymatic antioxidant) all protect against free radical damage.

13. Superoxide dismutase catalyzes which of the following reactions?

- A. $2\text{O}_2^- + 2\text{H}^+ \rightarrow \text{H}_2\text{O}_2 + \text{O}_2$
- B. $\text{H}_2\text{O}_2 + \text{O}_2^- \rightarrow \text{OH}\cdot + \text{O}_2$
- C. $\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \frac{1}{2}\text{O}_2$
- D. $\text{O}_2^- + \text{H}_2\text{O}_2 + \text{H}^+ \rightarrow 2\text{H}_2\text{O} + \text{O}_2$

Correct Answer: A. $2\text{O}_2^- + 2\text{H}^+ \rightarrow \text{H}_2\text{O}_2 + \text{O}_2$

Biochemical Rationale:

Superoxide dismutase (SOD) catalyzes the dismutation of two superoxide radicals into hydrogen peroxide and molecular oxygen. This reaction is a key antioxidant defense mechanism. Option C describes catalase activity, and Option D describes peroxidase activity.

14. When p53 increases in response to DNA damage, which of the following events occurs?

- A. p53 induces transcription of cdk4
- B. p53 induces transcription of cyclin D
- C. p53 induces E2F to activate S-phase genes
- D. p53 induces transcription of p21

Correct Answer: D. p53 induces transcription of p21

Biochemical Rationale:

p53 acts as a transcription factor that upregulates p21 (a cyclin-dependent kinase inhibitor) in response to DNA damage. p21 inhibits Cyclin-CDK complexes, preventing Rb phosphorylation and causing cell cycle arrest at the G1/S checkpoint to allow DNA repair.

15. Mutations in the p53 gene are the most common aberrations in spontaneous human cancers. The normal p53 protein affects the cell cycle by:

- A. Inducing cell cycle arrest and apoptosis in response to DNA damage
- B. Inducing phosphorylation of Rb in response to mitogens
- C. Directly inhibiting Cdk inhibitors in response to growth-inhibiting signals

- D. Increasing the activity of cyclin-dependent kinases by phosphorylation
- E. Enhancing telomerase activity during DNA repair

Correct Answer: A. Inducing cell cycle arrest and apoptosis in response to DNA damage

Biochemical Rationale:

Normal p53 functions as the "guardian of the genome." Upon DNA damage, it induces cell cycle arrest (via p21) to allow repair, or triggers apoptosis (via Bax, PUMA) if damage is irreparable. Loss of p53 function leads to unchecked proliferation and cancer.

16. Which one of the following is NOT true about the M check-point?

- A. Anaphase-promoting complex/cyclosome (APC/C) is a key regulator
- B. It is triggered mainly by activation of specific cyclin-Cdk complexes via phosphorylation
- C. Progression from metaphase to anaphase is triggered by cyclical proteolysis
- D. The spindle assembly checkpoint ensures proper chromosome attachment before separation
- E. Cohesin degradation is required for anaphase to proceed

Correct Answer: B. It is triggered mainly by activation of specific cyclin-Cdk complexes via phosphorylation

Biochemical Rationale:

The M checkpoint (spindle assembly checkpoint) is regulated primarily by the APC/C, which triggers proteolysis of securin and cyclin B. Activation of Cyclin B-CDK1 via phosphorylation/dephosphorylation triggers entry into mitosis (G2/M transition), not the M checkpoint itself.

17. A patient visited the emergency with methanol poisoning, and the physician started the treatment with administration of ethanol. High concentrations of ethanol reduce the effect of methanol poisoning by competitively inhibiting an enzyme known as.....?

Correct Answer: Alcohol dehydrogenase

Biochemical Rationale:

Ethanol acts as a competitive substrate for alcohol dehydrogenase, having a much higher affinity (lower K_m) than methanol. By occupying the enzyme's active site, ethanol prevents the metabolism of methanol to its toxic metabolites (formaldehyde and formic acid).

18. Organophosphates are used as pesticides which bind covalently to serine residue of active site of the enzyme acetylcholinesterase, and it is used as parasympathomimetic drug....?

Correct Answer: Acetylcholinesterase

Biochemical Rationale:

Organophosphates irreversibly inhibit acetylcholinesterase by phosphorylating the serine residue at the active site. This leads to accumulation of acetylcholine at synapses, causing excessive cholinergic stimulation (parasympathomimetic effects).

19. Name the enzyme obtained from *T. aquaticus*, which is used in PCR because of its thermostable character.....?

Correct Answer: Taq polymerase

Biochemical Rationale:

Taq polymerase is a thermostable DNA polymerase isolated from *Thermus aquaticus*. It can withstand the high temperatures (94–95°C) required for DNA denaturation during PCR cycles without losing activity.

20. The enzyme Xanthine oxidase is competitively inhibited by a molecule used as a drug for the treatment of gout ...?

Correct Answer: Allopurinol

Biochemical Rationale:

Allopurinol is a structural analog of hypoxanthine that competitively inhibits xanthine oxidase. This reduces uric acid production, lowering serum urate levels and preventing gout attacks.

21. Current prospective data show the highest risk of what type of cancer in women with pathogenic variants in the BRCA1 or BRCA2 genes?

- A. Adrenal cortical carcinoma
- B. Breast cancer
- C. Ovarian cancer
- D. Renal cell carcinoma

Correct Answer: B. Breast cancer

Biochemical Rationale:

BRCA1 and BRCA2 mutations confer a significantly elevated lifetime risk of breast cancer (up to 80%) and ovarian cancer. However, breast cancer is the most common malignancy associated with these mutations.

22. Which human disease is caused by trisomy 18?

- A. Down syndrome
- B. Patau syndrome
- C. Edwards syndrome
- D. Turner syndrome

Correct Answer: C. Edwards syndrome

Biochemical Rationale:

Edwards syndrome is caused by trisomy 18 (47,XX/XY,+18). It is characterized by severe intellectual disability, growth retardation, cardiac defects, and characteristic hand/foot abnormalities.

23. Which of the following is a result of trisomy 13?

- A. Down syndrome
- B. Patau syndrome
- C. Edwards syndrome
- D. Turner syndrome

Correct Answer: B. Patau syndrome

Biochemical Rationale:

Patau syndrome is caused by trisomy 13 (47,XX/XY,+13). It presents with severe craniofacial defects (cleft lip/palate), polydactyly, holoprosencephaly, and cardiac anomalies.

24. Klinefelter syndrome is characterized by which karyotype?

- A. 45,X (Turner syndrome)
- B. 47,XXX
- C. 47,XXY
- D. 47,YYY

Correct Answer: C. 47,XXY

Biochemical Rationale:

Klinefelter syndrome results from an extra X chromosome in males (47,XXY). Features include tall stature, gynecomastia, small testes, and infertility due to azoospermia.

25. Turner syndrome is caused by:

- A. Trisomy of chromosome 21
- B. Monosomy of the X chromosome
- C. Trisomy of chromosome 18
- D. Trisomy of chromosome 13

Correct Answer: B. Monosomy of the X chromosome

Biochemical Rationale:

Turner syndrome is caused by complete or partial monosomy of the X chromosome (45,X). It is the only viable whole-chromosome monosomy in humans, presenting with short stature, webbed neck, and ovarian dysgenesis.

26. Which of the following is an example of aneuploidy?

- A. Down syndrome
- B. Cystic fibrosis
- C. Sickle-cell anemia
- D. Hemophilia

Correct Answer: A. Down syndrome

Biochemical Rationale:

Aneuploidy refers to an abnormal number of chromosomes. Down syndrome (trisomy 21) is a classic example. Cystic fibrosis, sickle-cell anemia, and hemophilia are single-gene disorders, not chromosomal number abnormalities.

27. A newborn female (mother 36y, father 47y) has a round face, low hairline, hypertelorism, epicanthal folds, up-slanting palpebral fissures, long philtrum, high-arched palate, short/webbed neck, small hands and feet, clinodactyly of 5th fingers, overlapping toes, and diastasis of 1st and 2nd toes. No family history. Which of the following procedures is recommended to establish a diagnosis?

- A. Linkage analysis
- B. Expression analysis
- C. Gene sequencing analysis
- D. Karyotyping

Correct Answer: D. Karyotyping

Biochemical Rationale:

The clinical features described are suggestive of a chromosomal abnormality (likely Down

syndrome or another trisomy). Karyotyping is the standard diagnostic procedure to visualize chromosome number and structure.

28. Turner syndrome:

- A. Is the only whole-chromosome monosomy seen in live births in humans
- B. The most common congenital heart disease is bicuspid aortic valve / coarctation of the aorta
- C. Is milder in boys
- D. Girls with this condition are usually tall for age

Correct Answer: A. Is the only whole-chromosome monosomy seen in live births in humans

Biochemical Rationale:

Turner syndrome (45,X) is the only complete monosomy compatible with life. It affects females only (not boys), causes short stature (not tall), and is associated with cardiac defects including bicuspid aortic valve and coarctation, but option A is the most definitively true statement.

29. You are studying the effects of a potential pharmaceutical compound on your enzyme of interest. The addition of the compound results in an increase in K_m but does not affect V_{max} . What type of inhibition is this?

- A. Competitive inhibition
- B. Noncompetitive inhibition
- C. Peptide inhibition
- D. Uncompetitive inhibition

Correct Answer: A. Competitive inhibition

Biochemical Rationale:

Competitive inhibitors bind to the active site and can be overcome by increasing substrate concentration. This increases the apparent K_m (lower affinity) while V_{max} remains unchanged. Noncompetitive inhibition decreases V_{max} without affecting K_m .

30. A 56-year-old man recently returned from an overseas trip and reports severe substernal chest pain 3 days ago. Which of the following is the most appropriate laboratory test to order?

- A. Aspartate aminotransferase (AST)
- B. Creatine kinase, MB fraction (CK-MB)
- C. Lactate dehydrogenase, LD1 fraction
- D. Troponin I

Correct Answer: D. Troponin I

Biochemical Rationale:

Cardiac troponin I is the most sensitive and specific biomarker for myocardial infarction. It remains elevated for 7–10 days after onset, making it ideal for diagnosis 3 days post-event. CK-MB normalizes within 2–3 days, and AST/LDH are non-specific.

31. Which of the following best describes the mechanism by which allosteric effectors influence enzyme activity?

- A. Binding at the catalytic site
- B. Changing the conformation of the enzyme

- C. Forming high-energy complexes with substrate
- D. Increasing hydration of the active site
- E. Protecting against enzyme degradation

Correct Answer: B. Changing the conformation of the enzyme

Biochemical Rationale:

Allosteric effectors bind to regulatory sites (allosteric sites) distinct from the active site, inducing conformational changes that alter enzyme activity. This can either enhance (positive effector) or inhibit (negative effector) catalytic function.

32. Which of the following is an example of a proteolytic enzyme?

- A. Bromelain and papain
- B. Cellulase and myrosinase
- C. Maltase and diastase
- D. Phosphatase and amylase

Correct Answer: A. Bromelain and papain

Biochemical Rationale:

Bromelain (from pineapple) and papain (from papaya) are proteolytic enzymes (proteases) that hydrolyze peptide bonds. Cellulase degrades cellulose, maltase hydrolyzes maltose, and amylase breaks down starch.

33. The catalytic efficiency of two different enzymes can best be compared by examining:

- A. The rate of product formation
- B. The K_m value
- C. The k_{cat}/K_m ratio
- D. The molecular size of the enzyme

Correct Answer: C. The k_{cat}/K_m ratio

Biochemical Rationale:

The k_{cat}/K_m ratio (catalytic efficiency) reflects how efficiently an enzyme converts substrate to product at low substrate concentrations. It combines turnover number (k_{cat}) and substrate affinity (K_m), providing the best measure for comparing enzyme efficiency.

34. Cocaine esterase (an artificially synthesized analog of the transition state) degrades cocaine in the body. This is an example of:

- A. Abzyme
- B. Metalloenzyme
- C. Coenzyme
- D. All of the above

Correct Answer: A. Abzyme

Biochemical Rationale:

Abzymes (antibody enzymes) are catalytic antibodies designed to stabilize the transition state of a reaction. Cocaine esterase is an artificial enzyme developed using transition state analogs to catalyze cocaine hydrolysis.

35. Which of the following is a defining property of allosteric enzymes?

- A. They are inactive in the absence of accessory proteins
- B. They can bind more than one substrate simultaneously
- C. They can convert multiple different substrates to products
- D. They possess binding sites for regulatory molecules in addition to substrate binding
- E. They require additional cofactor binding sites for substrate binding

Correct Answer: D. They possess binding sites for regulatory molecules in addition to substrate binding

Biochemical Rationale:

Allosteric enzymes have distinct regulatory sites (allosteric sites) where effector molecules bind, in addition to their active sites. Binding of effectors induces conformational changes that modulate enzyme activity.

36. A hepatocyte that is actively participating in the cell cycle is observed to increase in size and to duplicate its organelles during a particular phase. In which phase does this hepatocyte presently reside?

- A. G1 phase
- B. G2 phase
- C. Prophase
- D. S phase

Correct Answer: B. G2 phase

Biochemical Rationale:

During the G2 phase, the cell continues to grow, synthesizes proteins necessary for mitosis, and duplicates organelles in preparation for cell division. DNA replication occurs in S phase, and G1 is the initial growth phase.

37. An actively cycling lymphocyte is observed to have a separation of the paired kinetochores and a migration of its sister chromatids toward opposite poles of the spindle. This mitotic lymphocyte presently resides in:

- A. Anaphase
- B. Metaphase
- C. Prometaphase
- D. Prophase

Correct Answer: A. Anaphase

Biochemical Rationale:

Anaphase is characterized by the separation of sister chromatids (via centromere cleavage) and their migration toward opposite spindle poles. Metaphase involves chromosome alignment at the metaphase plate, and prophase involves chromosome condensation.

38. Which of the following will increase the risk of the uncontrolled, abnormal growth of cells or disease in the body?

- A. Increase in the activity of DNA repair enzymes
- B. Decrease in the rate of mutations of protooncogenes
- C. Decrease in the activity of tumor suppressor genes

D. Increase in the activity of carcinogen-metabolizing enzymes

Correct Answer: C. Decrease in the activity of tumor suppressor genes

Biochemical Rationale:

Tumor suppressor genes (e.g., p53, Rb) normally inhibit cell proliferation and promote apoptosis. Loss or decreased activity of these genes removes critical brakes on cell division, leading to uncontrolled growth and cancer.

For questions 39–44, please select the letter that has the correct word/phrase that matches with the statements in the questions.

A. M-check point

B. SOD

C. Cyclin-Dependent Protein Kinases

D. Proto-oncogenes

E. Nitric oxide

F. G1-S check point

G. Tumor suppressor genes

H. Lycopene

I. G2-M check point

39. All replicated DNA molecules should be scanned for replication error if any ____

Correct Answer: I. G2-M check point

Biochemical Rationale:

The G2-M checkpoint ensures that DNA replication is complete and checks for any replication errors or DNA damage before the cell enters mitosis.

40. Integrity of DNA is checked before its replication ____

Correct Answer: F. G1-S check point

Biochemical Rationale:

The G1-S checkpoint verifies that DNA is intact and free of damage before committing the cell to DNA replication (S phase).

41. Alignment of every chromosome on spindle will be checked ____

Correct Answer: A. M-check point

Biochemical Rationale:

The M checkpoint (spindle assembly checkpoint) ensures that all chromosomes are properly attached to the spindle and aligned at the metaphase plate before anaphase begins.

42. Central component of cell-cycle control system ____

Correct Answer: C. Cyclin-Dependent Protein Kinases

Biochemical Rationale:

Cyclin-dependent protein kinases (CDKs) are the central regulators of cell cycle progression. They phosphorylate target proteins to drive cell cycle transitions when activated by cyclins.

43. Genes for signal transducer proteins ____

Correct Answer: D. Proto-oncogenes

Biochemical Rationale:

Proto-oncogenes encode proteins involved in signal transduction pathways that promote cell growth and division. When mutated or overexpressed, they become oncogenes that drive uncontrolled proliferation.

44. BRCA1, BRCA2; Retinoblastoma (Rb) genes and Ataxia telangiectasia (ATM) ____

Correct Answer: G. Tumor suppressor genes

Biochemical Rationale:

BRCA1, BRCA2, Rb, and ATM are all tumor suppressor genes. Their normal function is to inhibit cell proliferation, repair DNA damage, or promote apoptosis. Loss-of-function mutations in these genes predispose to cancer.

45. The first protein to cut during reparation of the damaged DNA by base excision repair system is:

- A. Peptidase
- B. Glycosylase
- C. AP endonuclease
- D. DNA ligase

Correct Answer: B. Glycosylase

Biochemical Rationale:

DNA glycosylase initiates base excision repair by recognizing and removing the damaged base through cleavage of the N-glycosidic bond, creating an AP site. AP endonuclease acts subsequently.

46. A 56-year-old man with chest pain is evaluated 3 days after onset. The most specific biomarker at this time is:

- A. AST
- B. CK-MB
- C. LDH1
- D. Troponin I

Correct Answer: D. Troponin I

Biochemical Rationale:

Cardiac troponin I remains elevated for 7–10 days after myocardial infarction, making it the most useful biomarker at 3 days post-onset. CK-MB normalizes within 48–72 hours, and AST/LDH are non-specific.

47. Which of the following biomolecules is the source of bioluminescence in fireflies?

- A. Lucigenin
- B. Luciferin
- C. Rhodopsin
- D. Retinal

Correct Answer: B. Luciferin

Biochemical Rationale:

Luciferin is the substrate that emits light when oxidized by luciferase in the presence of ATP and oxygen. This reaction produces the characteristic bioluminescence of fireflies.

48. Which protein is misfolded and infectious in nature, differing from most proteins by its biological function?

- A. Prion
- B. Thrombin
- C. Ricin
- D. Snake venom phospholipase

Correct Answer: A. Prion

Biochemical Rationale:

Prions are abnormally folded infectious proteins that cause neurodegenerative diseases (e.g., Creutzfeldt-Jakob disease, mad cow disease). They propagate by inducing conformational changes in normal prion proteins.

49. Which plasma protein deficiency enhances the Fenton reaction and free radical injury?

- A. Transferrin
- B. Ferritin
- C. Ceruloplasmin
- D. Albumin

Correct Answer: C. Ceruloplasmin

Biochemical Rationale:

Ceruloplasmin has ferroxidase activity, oxidizing Fe^{2+} to Fe^{3+} and preventing iron from participating in the Fenton reaction. Deficiency leads to increased free iron, enhanced hydroxyl radical generation, and oxidative tissue damage.

50. In mismatch repair, which enzyme identifies and excises the incorrect base?

- A. MutS/MutL complex
- B. DNA glycosylase
- C. DNA ligase I
- D. Telomerase

Correct Answer: A. MutS/MutL complex

Biochemical Rationale:

In mismatch repair, the MutS protein recognizes and binds to mismatched base pairs. MutL then activates downstream excision machinery to remove the incorrect nucleotide segment. DNA glycosylase functions in base excision repair, not mismatch repair.

51. The most common mutation found in spontaneous human cancers involves which gene?

- A. APC
- B. Rb
- C. p53
- D. BRCA1

Correct Answer: C. p53

Biochemical Rationale:

Mutations in the p53 tumor suppressor gene are the most common genetic alterations in human cancers, found in approximately 50% of all tumors. p53 is critical for cell cycle arrest, DNA

repair, and apoptosis.

52. Which of the following human diseases is caused by trisomy 18?

- A. Down syndrome
- B. Edwards syndrome
- C. Patau syndrome
- D. Turner syndrome

Correct Answer: B. Edwards syndrome

Biochemical Rationale:

Edwards syndrome is caused by trisomy 18 (47,XX/XY,+18), characterized by severe intellectual disability, growth retardation, and multiple congenital anomalies.

53. Which is the only complete monosomy compatible with life in humans?

- A. Monosomy 21
- B. Monosomy 18
- C. Monosomy X (Turner syndrome)
- D. Monosomy Y

Correct Answer: C. Monosomy X (Turner syndrome)

Biochemical Rationale:

Turner syndrome (45,X) is the only complete monosomy compatible with live birth in humans. All other complete autosomal monosomies are lethal in utero. Monosomy Y is not viable.

54. The microsatellites used in DNA fingerprinting for paternity testing are usually:

- A. Histocompatibility loci
- B. Centromeric repeats
- C. Short tandem repeats (STRs)
- D. Restriction enzyme cut sites

Correct Answer: C. Short tandem repeats (STRs)

Biochemical Rationale:

Short tandem repeats (STRs) are highly polymorphic microsatellite sequences (2–7 bp repeats) used as standard markers in DNA fingerprinting for paternity testing and forensic identification due to their high variability.

55. Mutations in mismatch repair genes (MLH1, MSH2, etc.) most commonly predispose to:

- A. Xeroderma pigmentosum
- B. Fanconi anemia
- C. Hereditary non-polyposis colorectal cancer (HNPCC)
- D. Bloom syndrome

Correct Answer: C. Hereditary non-polyposis colorectal cancer (HNPCC)

Biochemical Rationale:

HNPCC (Lynch syndrome) is caused by germline mutations in mismatch repair genes (MLH1, MSH2, MSH6, PMS2). Defective mismatch repair leads to microsatellite instability and

increased risk of colorectal and other cancers.

